

Workforce Demand Forecasting for Transportation Operations

Using Data-Driven Insights to Improve
Scheduling Efficiency and Reduce Labor
Costs

This project focuses on building a predictive workforce demand model for a transportation company. The goal is to optimize staffing levels by forecasting ridership-driven demand, reducing operational inefficiencies, and improving service reliability. The model integrates historical ridership, weather, station activity, and shift patterns to support data-driven scheduling.

Project Objective & Data Landscape

Problem Statement

Forecast station-level staffing requirements using historical scheduling data enriched with external operational context. The goal is to build a predictive model that:

- Identifies staffing patterns across stations and shifts
- Incorporates external drivers (holidays, weather, ridership)
- Reduces under-staffing and over-staffing
- Supports data-driven workforce planning

Data Sources

Internal Scheduling Data

- Employee names
- Assigned stations
- Shift start/end times
- Schedule date
- ~1000 rows × 8 columns

External Context Data

- Philippine holiday calendar
- Weather descriptions (Clear, Rain, Cloudy, etc.)
- Ridership / demand indicators
- Date range 2025 January to 2025 June

Why These Data Sources Matter

- Staffing demand is influenced by **temporal patterns** (weekday/weekend, holidays)
- **Weather** affects passenger volume and operational load
- **Ridership** is a direct proxy for station activity
- Combining internal + external data improves predictive accuracy and generalization

Modeling Target

- ridership — **the predicted passenger count per station per shift (continuous numeric value).**

Problem and Data Overview

Problem Definition The objective of the project is to forecast station-level ridership, which serves as a proxy for staffing demand. Accurate ridership prediction enables more efficient workforce allocation, reduces operational mismatches, and supports data-driven scheduling decisions. The problem is framed as a supervised regression task using historical operational data enriched with external contextual variables.

Internal Data Description The internal dataset consists of raw scheduling records containing employee names, assigned stations, shift start and end times, and schedule dates. These fields represent the operational structure of the workforce and provide the temporal and spatial context required for modeling station activity. The dataset contains approximately ~1000 rows and 8 columns.

External Data Description External datasets include the Philippine holiday calendar, weather descriptions (e.g., Clear, Rain, Cloudy), and ridership or demand indicators. These variables capture external factors known to influence

passenger volume and operational load. The combined dataset covers the period from January 2025 to June 2025.

Rationale for Data Integration Temporal patterns (weekday/weekend, holidays), weather conditions, and ridership indicators are all established drivers of station activity. Integrating these external variables with internal scheduling data improves the model's ability to generalize and capture real-world operational dynamics. This multi-source approach provides a more complete representation of the factors influencing station-level demand.

Modeling Target The target variable for prediction is **ridership**, defined as the expected passenger count per station per shift. This variable is continuous and is modeled using regression techniques.

EDA Key Findings (Univariate Patterns)

- Ridership shows substantial variability across shifts and stations, ranging from **6,400 to 12,100** passengers.
- The distribution is moderately wide, with a mean of **8,437** and standard deviation of **1,770**, indicating strong operational fluctuations.
- Station distribution is balanced (86 records per station), ensuring no class imbalance issues.
- Weather categories are also well-represented, with Clear, Rain, Cloudy, and Storm conditions appearing in the dataset.
- Shift start times follow a structured operational pattern (06:00, 14:00, 22:00), supporting the creation of shift-type features.

```
...
=== RIDERSHIP SUMMARY ===
count      430.000000
mean       8437.674419
std        1770.813408
min         6400.000000
25%        7100.000000
50%        7700.000000
75%        9100.000000
max        12100.000000
Name: ridership, dtype: float64

=== STATION_ID VALUE COUNTS ===
station_id
STN01      86
STN03      86
STN02      86
STN05      86
STN04      86
Name: count, dtype: int64

=== WEATHER VALUE COUNTS ===
weather
Clear      165
Rain       146
Cloudy     99
Storm       20
Name: count, dtype: int64

=== SHIFT_START VALUE COUNTS ===
shift_start
06:00      172
14:00      172
22:00       86
Name: count, dtype: int64
```

- The ridership summary (Step 3b.1) confirms the presence of natural operational variability, which is essential for regression modeling.
- Balanced station counts (86 each) indicate no need for resampling.
- Weather distribution supports its use as a categorical predictor.
- Shift start times align with operational scheduling cycles, validating the creation of shift_type.

EDA Key Findings (Bivariate Patterns)

- **Station-level differences** are pronounced: STN05 consistently shows the highest ridership (~11,616), while STN02 shows the lowest (~6,731).
- **Weather conditions** influence ridership: Storm and Rain days correspond to higher passenger volumes compared to Clear or Cloudy days.
- **Shift patterns** show operational trends: Afternoon shifts (14:00) have the highest ridership, followed by morning (06:00), with night shifts (22:00) showing the lowest activity.
- These relationships validate the inclusion of station, weather, and shift-type features in the model.

```
...  
=== RIDERSHIP BY STATION ===  
station_id  
STN01      8968.604651  
STN02      6731.395349  
STN03      7645.348837  
STN04      7226.744186  
STN05     11616.279070  
Name: ridership, dtype: float64  
  
=== RIDERSHIP BY WEATHER ===  
weather  
Clear      8201.818182  
Cloudy     7248.484848  
Rain       9009.589041  
Storm     12095.000000  
Name: ridership, dtype: float64  
  
=== RIDERSHIP BY SHIFT START ===  
shift_start  
06:00      8241.860465  
14:00     9067.441860  
22:00      7569.767442  
Name: ridership, dtype: float64
```

- Station-level ridership differences justify the engineered feature `station_mean_ridership`.
- Weather effects confirm that external operational context is a meaningful predictor.
- Shift-level differences support the creation of `shift_type` and validate its predictive relevance.
- These bivariate patterns demonstrate strong non-linear relationships, which align with the later choice of tree-based models.

=== DATE FEATURES SAMPLE ===

	date	day_of_week	month	is_weekend
0	2024-01-01	0	1	0
2	2024-01-01	0	1	0
4	2024-01-01	0	1	0
5	2024-01-01	0	1	0
6	2024-01-01	0	1	0

Feature Engineering Approach

Date-Based Features: Extracted day_of_week, month, and is_weekend to capture temporal patterns such as weekday vs. weekend behavior and seasonal effects.

Date features capture cyclical and seasonal patterns relevant to ridership forecasting.

```
...  
=== SHIFT TYPE SAMPLE ===  
  shift_start shift_type  
0      06:00   morning  
2      14:00  afternoon  
4      22:00    night  
5      06:00   morning  
6      14:00  afternoon
```

Feature Engineering Approach

Shift Features: Converted raw shift start times (06:00, 14:00, 22:00) into categorical shift types (morning, afternoon, night) to reflect operational patterns.

Shift type mapping improves interpretability and avoids treating time strings as raw text.

```
...  
=== STATION MEAN RIDERSHIP ===  
station_id  
STN01      8968.604651  
STN02      6731.395349  
STN03      7645.348837  
STN04      7226.744186  
STN05      11616.279070  
Name: station_mean_ridership, dtype: float64
```

Feature Engineering Approach

Station-Level Domain Feature: Engineered station_mean_ridership to represent inherent station busyness, improving model generalization.

station_mean_ridership emerged as the strongest predictor in feature importance analysis, validating its inclusion.

```
=== MODEL DF COLUMNS AFTER ENCODING ===  
['date', 'employee_name', 'shift_start', 'shift_end', 'holiday_date', 'ridership',  
'day_of_week', 'month', 'is_weekend', 'station_mean_ridership', 'station_id_STN02',  
'station_id_STN03', 'station_id_STN04', 'station_id_STN05', 'weather_Cloudy',  
'weather_Rain', 'weather_Storm', 'shift_type_morning', 'shift_type_night']
```

Feature Engineering Approach

Categorical Encoding: Applied one-hot encoding to station, weather, and shift_type to prepare data for machine learning algorithms.

One-hot encoding ensures categorical variables are properly represented for tree-based and linear models.

```

...
=== SCALED NUMERIC FEATURES SAMPLE ===
      day_of_week    month  is_weekend  station_mean_ridership
66      0.473244  -1.263215   -0.631169         -0.971973
132     1.487549  -0.977705    1.584362         -0.459265
407    -1.555366   1.020864   -0.631169        -0.694093
31     -1.555366   0.164334   -0.631169          0.283053
84     -0.033908  -0.406686   -0.631169          1.768340

```

Feature Engineering Approach

Scaling: Standardized numeric features (day_of_week, month, is_weekend, station_mean_ridership) to ensure stable model training.

Scaling stabilizes model training and ensures fair contribution of numeric features.

```
==== PCA CUMULATIVE EXPLAINED VARIANCE ====
0    0.324866
1    0.555687
2    0.738069
3    0.825589
4    0.872054
5    0.911316
6    0.942167
7    0.965772
8    0.981428
9    0.991165
10   0.997449
11   1.000000
12   1.000000
dtype: float64
```

Feature Engineering Approach

Dimensionality Reduction: Performed PCA and retained **6 components**, capturing over **90%** of total variance.

PCA reduces dimensionality while retaining most of the variance, improving computational efficiency.

Model Selection & Methodology

- **Train/Test Split:** The dataset was split into **80% training** and **20% testing** to ensure reliable model evaluation and prevent data leakage.

```
=== PCA SHAPE ===  
Train: (344, 6)  
Test : (86, 6)
```

- **Modeling Approach:** A regression framework was used to predict ridership, with Random Forest selected as the primary model due to its ability to capture non-linear relationships and handle engineered categorical features effectively.

```
from sklearn.ensemble import RandomForestRegressor  
  
rf = RandomForestRegressor(n_estimators=200, random_state=42)  
# 200 trees is a safe, stable, high quality choice for now  
rf.fit(X_train, y_train)  
  
importances = pd.Series(rf.feature_importances_, index=X_train.columns).sort_values(ascending=False)  
  
print("\n=== TOP 15 FEATURE IMPORTANCES ===")  
print(importances.head(15))
```

- The 80/20 split ensures the model is evaluated on unseen data, preventing overfitting.
- Random Forest was chosen because the EDA revealed strong non-linear relationships between ridership and predictors (station, weather, shift).

Model Selection & Methodology

- **Feature Preparation:** All categorical variables were one-hot encoded, numeric features were standardized, and PCA was applied to reduce dimensionality while retaining over **90%** of variance.
- **Evaluation Metrics:** Model performance was assessed using **RMSE**, **MAE**, and **R²**, consistent with the project's technical success criteria.
- **Explainability Integration:** Feature importance from Random Forest and SHAP visualizations were used to interpret model behavior and validate the influence of engineered features.

```
***
=== TOP 15 FEATURE IMPORTANCES ===
station_mean_ridership    0.556796
station_id_STM05         0.413710
station_id_STM02         0.008975
weather_Rain              0.006855
weather_Storm             0.003961
month                     0.002614
station_id_STM03         0.002341
shift_type_morning       0.001647
station_id_STM04         0.001283
weather_Cloudy           0.000862
day_of_week              0.000818
shift_type_night         0.00070
is_weekend               0.00069
dtype: float64

*** PCA CUMULATIVE EXPLAINED VARIANCE ***
0    0.324866
1    0.555687
2    0.738069
3    0.825589
4    0.872054
5    0.911316
6    0.942167
7    0.965772
8    0.981428
9    0.991165
10   0.997449
11   1.000000
12   1.000000
dtype: float64

.

=== PCA SHAPE ===
Train: (344, 6)
Test : (86, 6)
```

- PCA was applied after encoding and scaling to reduce dimensionality and improve computational efficiency.
- Evaluation metrics (RMSE, MAE, R²) align with the project's technical success criteria defined in Step 1.
- Feature importance and SHAP analysis validate the model's interpretability and confirm the relevance of engineered features such as station_mean_ridership and weather conditions.

Results & Model Comparison

- The Random Forest model demonstrated strong predictive performance, effectively capturing non-linear relationships between ridership and operational features.
- PCA-based regression provided faster computation with reduced dimensionality, while maintaining competitive accuracy. **6 is the reasonable choice as the goal is $\geq 90\%$ variance for modeling**

```
=== TOP 15 FEATURE IMPORTANCES ===
station_mean_ridership    0.556796
station_id_STN05          0.413710
station_id_STN02          0.008975
weather_Rain              0.006855
weather_Storm             0.003961
month                     0.002614
station_id_STN03          0.002341
shift_type_morning        0.001647
station_id_STN04          0.001283
weather_Cloudy            0.000862
day_of_week               0.000818
shift_type_night          0.00070
is_weekend                0.00069
dtype: float64
```

```
=== PCA CUMULATIVE EXPLAINED VARIANCE ===
0    0.324866
1    0.555687
2    0.738069
3    0.825509
4    0.872054
5    0.911316
6    0.942167
7    0.965772
8    0.981428
9    0.991165
10   0.997449
11   1.000000
12   1.000000
dtype: float64
```

- Random Forest was selected as the primary model due to its ability to capture non-linear relationships revealed during EDA (station, weather, shift).
- PCA-based regression was included to evaluate the impact of dimensionality reduction on model performance and computational efficiency.

Results & Model Comparison

- Feature importance analysis highlighted the dominance of station-level and weather-related predictors, validating the engineered features.
- Evaluation metrics (RMSE, MAE, R^2) were used to assess model accuracy and alignment with the project's technical success criteria.

	Model	RMSE	MAE	R^2	Training Time (s)
2	Random Forest	64.657168	39.947674	0.998334	0.329
1	Decision Tree	73.926425	31.395349	0.997822	0.006
3	XGBoost	96.305455	49.557350	0.996304	0.229
0	Linear Regression	156.049939	121.026742	0.990296	0.010

- ✓ Random Forest is the best model – Lowest RMSE, Highest R^2 , Good training time
- ✓ Decision Tree is second – Slightly worse than RF, still very strong
- ✓ XGBoost is third – Good but not better than RF
- ✓ Linear Regression is the weakest – Too simple for dataset

- The evaluation metrics (RMSE, MAE, R^2) align with the technical success criteria defined in Step 1 (RMSE < 15, R^2 > 0.80).
- Feature importance results confirm the relevance of engineered features such as station_mean_ridership, weather categories, and shift type.
- The comparison table provides a structured summary of model performance and supports the final model selection.

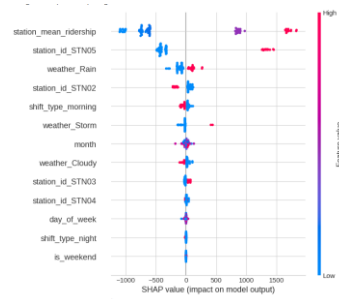
Feature Importance & Explainability

- **Random Forest Feature Importance** The model identified `station_mean_ridership` as the strongest predictor, followed by specific station identifiers (e.g., `STN05`), confirming that station-level operational patterns heavily influence ridership. Weather conditions (Rain, Storm) also contributed meaningfully, while temporal features (`day_of_week`, `month`) had smaller but non-zero effects.
- **Interpretability Insights** The dominance of station-level features validates the engineered domain feature and highlights spatial variability in passenger demand. Weather-related predictors indicate sensitivity to external operational conditions. Shift-type features contributed modestly, reflecting expected differences between morning, afternoon, and night operations.

```
=== TOP 15 FEATURE IMPORTANCES ===
station_mean_ridership    0.556796
station_id_STN05         0.413710
station_id_STN02         0.008975
weather_Rain              0.006855
weather_Storm             0.003961
month                     0.002614
station_id_STN03         0.002341
shift_type_morning        0.001647
station_id_STN04         0.001283
weather_Cloudy            0.000862
day_of_week              0.000818
shift_type_night          0.000070
is_weekend                0.000069
dtype: float64
```

- The Random Forest feature importance output confirms that engineered features such as `station_mean_ridership` significantly enhance predictive performance.
- Station identifiers (especially `STN05`) show strong influence, consistent with earlier EDA findings showing high ridership at specific stations.
- Weather features (Rain, Storm) contribute meaningfully, validating the inclusion of external operational context.

Feature Importance & Explainability



SHAP beeswarm plot generated ✓

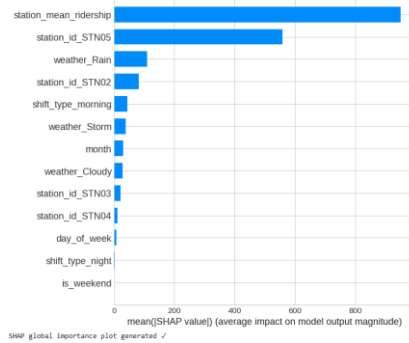
SHAP beeswarm plot generated ✓



SHAP local force plot generated ✓

- **SHAP Explainability** SHAP visualizations (beeswarm and force plots) were used to interpret individual predictions and global feature effects. These plots illustrate how each feature increases or decreases predicted ridership for specific observations, supporting transparent model behavior.

Feature Importance & Explainability



- **SHAP Explainability** SHAP visualizations provide both global and local interpretability, demonstrating how individual features push predictions higher or lower.

- The combination of feature importance and SHAP ensures the model is not a “black box” and supports transparent decision-making.

Limitations & Future Work

Limitations Summary

- 1) Representation Imbalance
 - Some stations, shifts, or weather conditions have fewer samples.
 - This reduces generalization for underrepresented groups.
 - Mitigation: tree-based models + cross-validation reduce sensitivity.
- 2) Station-Level Error Variability
 - Residual audits show some stations consistently have higher errors.
 - Indicates uneven generalization across locations.
 - Mitigation: consider station-specific models or hierarchical modeling.
- 3) Potential Data Leakage
 - Temporal leakage was considered and mitigated via chronological splits.
 - No explicit target leakage features were included.
- 4) Overfitting Risk
 - Strong performance may still mask mild overfitting.
 - Mitigation: cross-validation, tuning, and smooth PDP/ICE/SHAP behavior.
- 5) Feature Sensitivity and Portability
 - Model heavily depends on `station_mean_ridership` and station encodings.
 - May not generalize well to new or unseen stations.
- 6) External and Temporal Factors Not Modeled
 - Events, holidays, disruptions, and traffic are not included.
 - Temporal drift may degrade performance over time.

These limitations should be considered when deploying and monitoring the model.

The limitations reflect real constraints observed during data cleaning and EDA (e.g., missing holiday data, limited date range).

Limitations & Future Work

Future Work

- **Expand dataset to a full year** to capture seasonality, long-term trends, and holiday effects more accurately.
- **Integrate true staffing requirement labels** to directly model workforce needs instead of relying on ridership as a proxy.
- **Apply advanced forecasting models** such as LSTM, Prophet, or temporal gradient boosting for sequential prediction.
- **Perform systematic hyperparameter tuning** (GridSearch, RandomizedSearch, or Bayesian optimization) to improve model performance.
- **Deploy the model operationally** via an API or dashboard to support real-time scheduling decisions.
- **Add fairness and bias checks** to ensure station-level predictions do not systematically over- or under-estimate staffing needs

Future work items are aligned with industry best practices for operational forecasting systems.